```sql
# 2. Write a SQL statement to create a simple table countries including columns country_id,country_name and region_id which is already exists.
CREATE TABLE IF NOT EXISTS countries (
    country_id INT,
    country_name VARCHAR(50),
    region_id INT
);
SELECT * FROM COUNTRIES;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country_id	country_name	region_id

```sql
#3. Write a SQL statement to create the structure of a table dup_countries similar to countries.
CREATE TABLE dup_countries LIKE countries;
select * FROM dup_countries;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country_id	country_name	region_id

```sql
#/*4. Write a SQL statement to create a duplicate copy of countries table including structure and data by name dup_countries
CREATE TABLE dup_countries AS SELECT * FROM countries;
select * from dup_countries;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country_id	country_name	region_id

```sql
#5. Write a SQL statement to create a table countries set a constraint NULL.
CREATE TABLE countries (
    country_id INT NULL,
    country_name VARCHAR(50) NULL,
    region_id INT NULL
);
select * from countries;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country_id	country_name	region_id

```sql
#6. Write a SQL statement to create a table named jobs including columns job_id, job_title, min_salary, max_salary and check whether the max_salary amount exceeding the
CREATE TABLE jobs (
    job_id INT PRIMARY KEY,
    job_title VARCHAR(50),
    min_salary DECIMAL(10,2),
    max_salary DECIMAL(10,2),
    CHECK (max_salary <= 25000));
select * from jobs;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

job_id	job_title	min_salary	max_salary
NULL	NULL	NULL	NULL

```sql
/* 7. Write a SQL statement to create a table named countries including columns country_id, country_name and region_id and make
sure that no countries except Italy, India and China will be entered in the table.*/
create table countries(
    countri_id int primary key,
    country_name varchar(90),
    region_id enum('india','china','italy'));
select * from countries;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

countri_id	country_name	region_id
NULL	NULL	NULL

```sql
/* 8. Write a SQL statement to create a table named job_histry including columns employee_id, start_date, end_date, job_id and department_id and make sure that the value against
column end_date will be entered at the time of insertion to the format like '--/--/----'.*/
CREATE TABLE job_histry (employee_id INT, start_date DATE,
    end_date VARCHAR(10),
    job_id INT,
    department_id INT);
desc job_histry;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES		NULL	
start_date	date	YES		NULL	
end_date	varchar(10)	YES		NULL	
job_id	int	YES		NULL	
department_id	int	YES		NULL	

Result Grid

Form Editor

```sql
/*9. Write a SQL statement to create a table named countries including columns country_id,country_name and region_id
and make sure that no duplicate data against column country_id will be allowed at the time of insertion.*/
CREATE TABLE countries (
    country_id INT UNIQUE,
    country_name VARCHAR(50),
    region_id INT);
select * from countries;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country_id	country_name	region_id

```sql
/*10.create a table named jobs including columns job_id, job_title, min_salary and max_salary, and make sure that, the default value for job_title is blank and
min_salary is 8000 and max_salary is NULL will be entered automatically at the time of insertion if no value assigned for the specified columns.*/
CREATE TABLE jobs (job_id INT,
    job_title VARCHAR(50) DEFAULT '',
    min_salary DECIMAL(10,2) DEFAULT 8000,
    max_salary DECIMAL(10,2) DEFAULT NULL));
    desc jobs;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
job_id	int	YES		NULL	
job_title	varchar(50)	YES			
min_salary	decimal(10,2)	YES		8000.00	
max_salary	decimal(10,2)	YES		NULL	

```sql
/*11.create a table named countries including columns country_id, country_name and region_id and make sure that the country_id column
 will be a key field which will not contain any duplicate data at the time of insertion.*/
 CREATE TABLE countries (
    country_id INT PRIMARY KEY,
    country_name VARCHAR(50),
    region_id INT);
desc countries;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
country_id	int	NO	PRI	NULL	
country_name	varchar(50)	YES		NULL	
region_id	int	YES		NULL	

```sql
/*12. create a table countries including columns country_id, country_name and region_id and make sure that the column country_id
will be unique and store an auto incremented value.*/
CREATE TABLE countries (
    country_id INT AUTO_INCREMENT UNIQUE,
    country_name VARCHAR(50),
    region_id INT);
desc countries;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
country_id	int	NO	PRI	NULL	auto_increment
country_name	varchar(50)	YES		NULL	
region_id	int	YES		NULL	

```sql
/*13. Write a SQL statement to create a table countries including columns country_id, country_name
 and region_id and make sure that the combination of columns country_id and region_id will be unique.*/
 CREATE TABLE countries (
    country_id INT,
    country_name VARCHAR(50),
    region_id INT,
    UNIQUE (country_id, region_id));
 desc countries;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
country_id	int	YES	MUL	NULL	
country_name	varchar(50)	YES		NULL	
region_id	int	YES		NULL	

```sql
# 14. create table job_history with foreign key
CREATE TABLE job_history (employee_id INT UNIQUE,
    start_date DATE, end_date DATE, job_id VARCHAR(10)
    department_id INT,
    FOREIGN KEY (job_id) REFERENCES jobs(job_id));
desc job_history;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES	UNI	NULL	
start_date	date	YES		NULL	
end_date	date	YES		NULL	
job_id	varchar(10)	YES	MUL	NULL	
department_id	int	YES		NULL	

```sql
#15. create table employee and make foreign key refrencing from table department
CREATE TABLE employees (employee_id INT UNIQUE,first_name VARCHAR(50),
    last_name VARCHAR(50),email VARCHAR(100),
    phone_number VARCHAR(20), hire_date DATE, job_id VARCHAR(10),
    salary DECIMAL(10,2), commission DECIMAL(10,2),
    manager_id DECIMAL(6,0), department_id DECIMAL(4,0),
    FOREIGN KEY (department_id, manager_id) REFERENCES departments(department_id, manager_id));
    desc employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES	UNI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
email	varchar(100)	YES		NULL	
phone_number	varchar(20)	YES		NULL	
hire_date	date	YES		NULL	
job_id	varchar(10)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
commission	decimal(10,2)	YES		NULL	
manager_id	decimal(6,0)	YES		NULL	
department_id	decimal(4,0)	YES	MUL	NULL	

```sql
#16.Create employee with two FOREIGN KEY constraints
CREATE TABLE employee (
    employee_id INT UNIQUE, first_name VARCHAR(50),last_name VARCHAR(50),
    email VARCHAR(100), phone_number VARCHAR(20),hire_date DATE, job_id VARCHAR(10),
    salary DECIMAL(10,2),commission DECIMAL(10,2),manager_id DECIMAL(6,0),
    department_id DECIMAL(4,0), FOREIGN KEY (department_id) REFERENCES departments(department_id),
     FOREIGN KEY (job_id) REFERENCES jobs(job_id)) ;
     desc employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES	UNI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
email	varchar(100)	YES		NULL	
phone_number	varchar(20)	YES		NULL	
hire_date	date	YES		NULL	
job_id	varchar(10)	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	
commission	decimal(10,2)	YES		NULL	
manager_id	decimal(6,0)	YES		NULL	
department_id	decimal(4,0)	YES	MUL	NULL	

```sql
#17.FOREIGN KEY with ON UPDATE CASCADE and ON DELETE RESTRICT
CREATE TABLE employeees (employee_id INT UNIQUE,first_name VARCHAR(50),
    last_name VARCHAR(50),job_id varchar(10),salary DECIMAL(10,2),
    FOREIGN KEY (job_id) REFERENCES jobs(job_id)
    ON UPDATE CASCADE ON DELETE RESTRICT) ;
    desc employeees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES	UNI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
job_id	varchar(10)	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	

```sql
#18.FOREIGN KEY with ON DELETE CASCADE and ON UPDATE RESTRICT
CREATE TABLE employees (employee_id INT UNIQUE,first_name VARCHAR(50),
    last_name VARCHAR(50), job_id varchar(10), salary DECIMAL(10,2),
    FOREIGN KEY (job_id) REFERENCES jobs(job_id)
    ON DELETE CASCADE ON UPDATE RESTRICT) ;
    desc employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES	UNI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
job_id	varchar(10)	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	

```sql
#Q19. FOREIGN KEY with ON DELETE SET NULL and ON UPDATE SET NULL
CREATE TABLE eemployees (
    employee_id INT UNIQUE,
    first_name VARCHAR(50),last_name VARCHAR(50),
    job_id VARCHAR(10) NULL, salary DECIMAL(10,2),
    FOREIGN KEY (job_id) REFERENCES jobs(job_id)
    ON DELETE SET NULL ON UPDATE SET NULL);
DESC EEMPLOYEES;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES	UNI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
job_id	varchar(10)	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	

```sql
#20.FOREIGN KEY with ON DELETE NO ACTION and ON UPDATE NO ACTION
CREATE TABLE employeess (
    employee_id INT UNIQUE,
    first_name VARCHAR(50),last_name VARCHAR(50),
    job_id VARCHAR(10), salary DECIMAL(10,2),
    FOREIGN KEY (job_id) REFERENCES jobs(job_id)
    ON DELETE NO ACTION ON UPDATE NO ACTION);
desc employeess;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	YES	UNI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
job_id	varchar(10)	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	